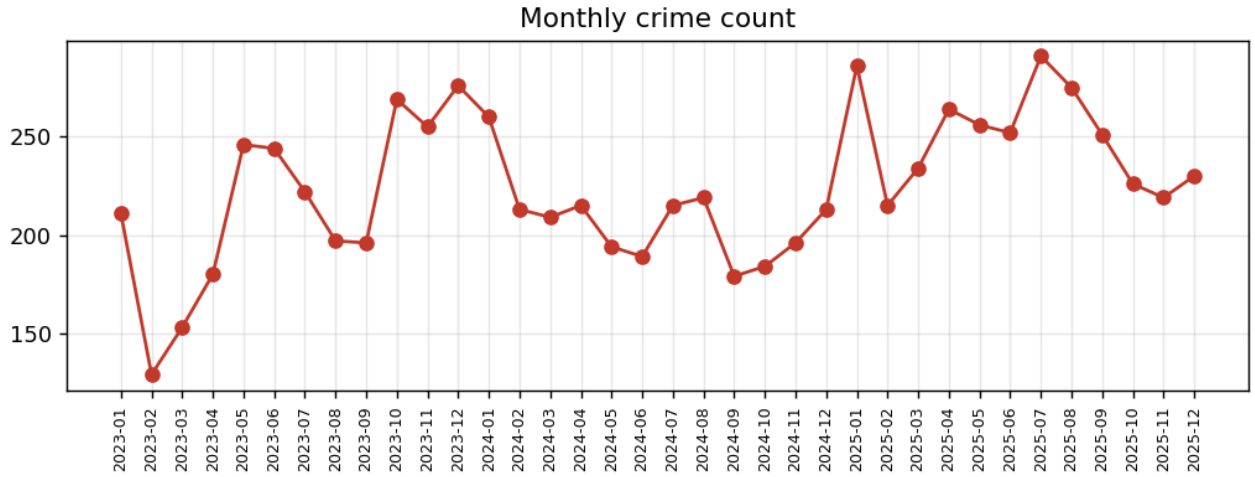
AI Crime Analysis Report

Generated 2026-10-13 12:47 | Data: 2023-01-01 to 2025-12-31 | 8063 records

1. Executive summary

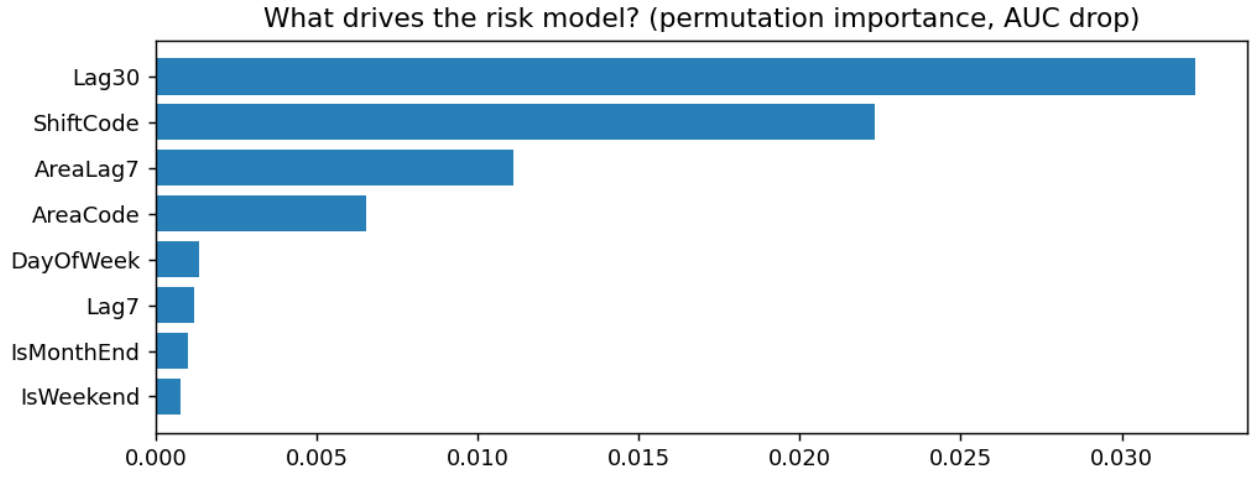
- Total records: 8063 across 9 areas. Most crimes: Market Commercial. Night shift share: 45%.
- Top hotspot: #1 near Market Commercial (1652 crimes, mostly Theft, peak: Night).
- Forecast: about 225 crimes expected city-wide in the next 30 days.
- Highest-risk grid cell next week: R3C2 (expected 6.4 crimes, main: Theft).
- Latest anomaly alert: Market Commercial: 20 crimes vs ~11.0 expected (+82%), mostly Theft (week of 2025-12-15).



2. Risk model (honest evaluation)

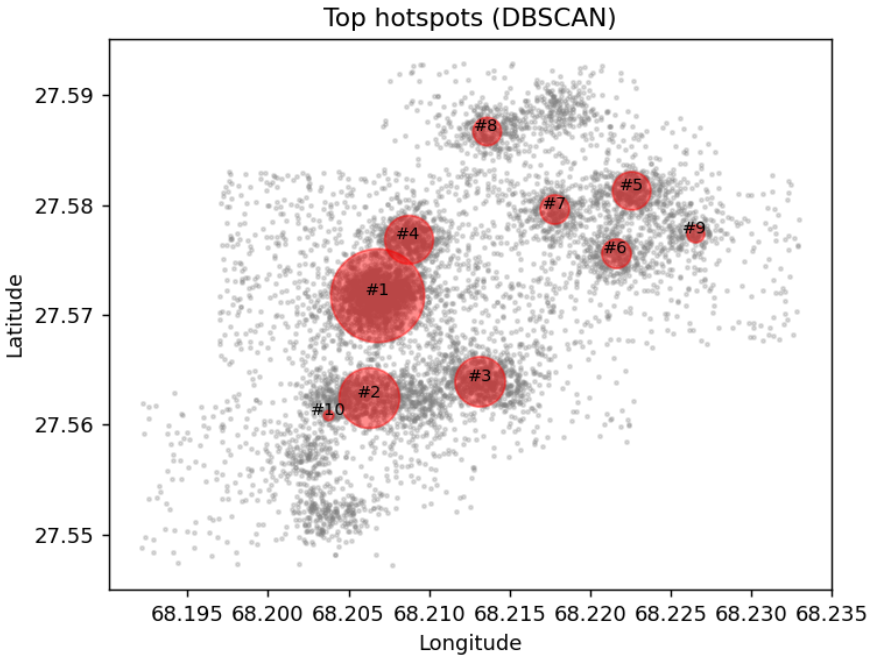
Target: will at least one crime occur in this area during this shift on this day? Features use only PAST information (lags), and the model is tested on the final 365 days (time-based split), so there is no data leakage.

Model	AUC	Accuracy	Precision	Recall	F1
Baseline (historical average)	0.63	0.61	0.34	0.57	0.42
RandomForest	0.66	0.62	0.35	0.61	0.44
GradientBoosting	0.65	0.62	0.35	0.58	0.43



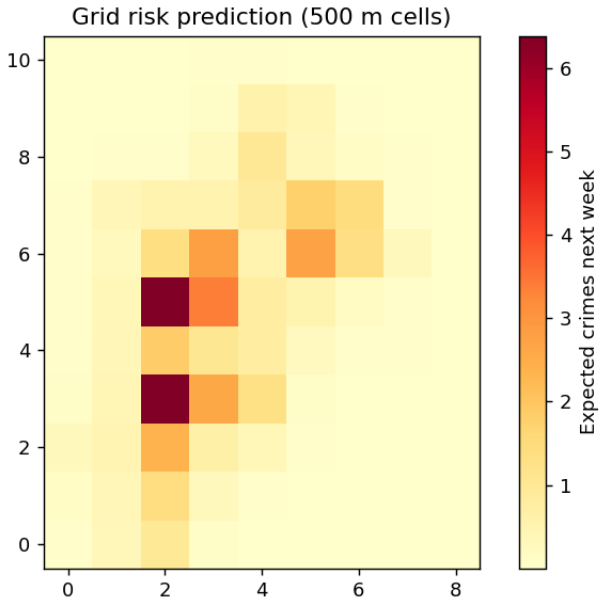
3. Hotspots (DBSCAN clustering)

Rank	Area	Crimes	Dominant_Crime	Peak_Shift	Last_90_Days
1	Market Commercial	1652	Theft	Night	124
2	Civil Line	689	Theft	Night	75
3	Market Residential	476	Theft	Night	24
4	Market Commercial	445	Theft	Night	39
5	Waleed Area	278	Theft	Night	18
6	Waleed Extension	162	Theft	Night	20

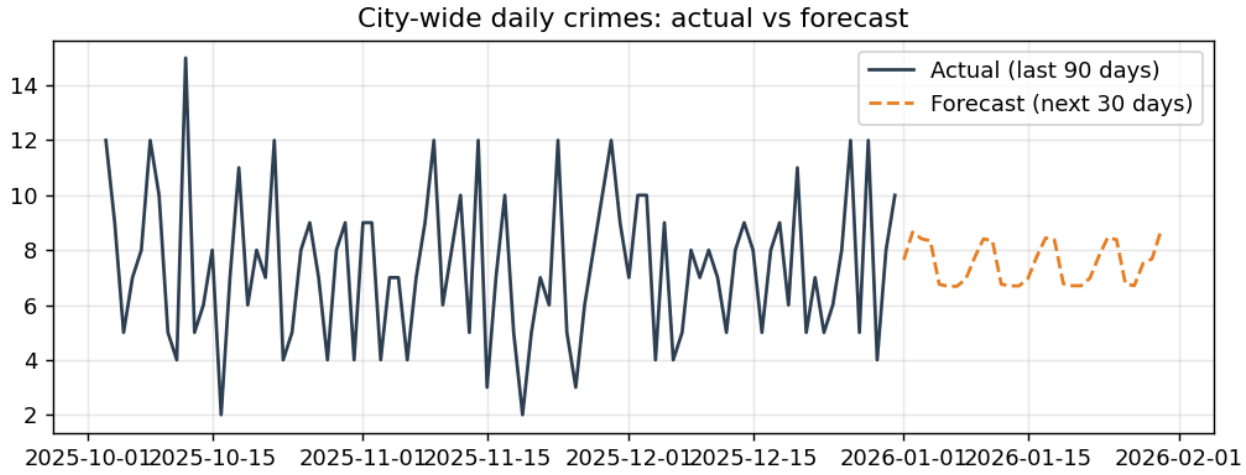


4. Grid risk prediction (next week)

Rank	Cell_ID	Pred_Next_Week	Last_4wk_Avg	Main_Crime	Risk_Level
1	R3C2	6.39	5.75	Theft	High
2	R5C2	6.36	4.50	Theft	High
3	R5C3	3.39	3.50	Theft	High
4	R6C3	2.79	3.25	Theft	High
5	R6C5	2.72	3.00	Theft	High
6	R3C3	2.62	2.75	Theft	High

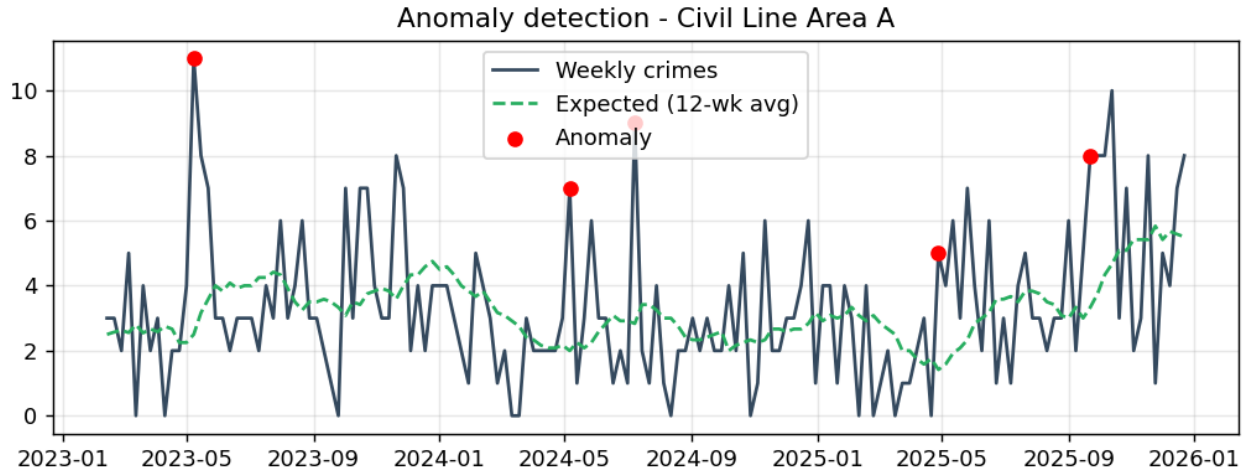

Grid risk prediction (500 m cells)
10
8
6
4
2
0
0
2
4
6
8
6
5
4
3
2
1
Expected crimes next week

5. Forecast (next 30 days)



Model	Daily_MAE	Weekly_MAE
AI model (Poisson boosting)	0.75	2.27
Baseline: last 28-day average	0.75	2.30
Baseline: area historical mean	0.75	2.34

6. Anomaly alerts



Week_Start	Area	Count	Expected	Z_Score	Level
2025-12-15	Market Commercial	20	11.00	2.52	High
2025-10-27	Market Commercial	17	10.30	2.07	Medium
2025-08-11	Waleed North	16	6.10	2.92	High
2025-07-28	Market Area	13	5.60	3.14	High
2025-06-16	Waleed Area	22	7.50	4.93	High
2025-04-07	Market Residential	14	5.00	3.22	High

7. Fairness / bias audit

- [WARNING] Market Area: false-alarm rate is 1.70x overall (NOT explained by a higher crime rate - investigate).
- [INFO] Market Commercial: false-alarm rate is 2.93x overall (expected: this area's real crime rate (0.44) is much higher than overall (0.25)).
- [INFO] Waleed Area: false-alarm rate is 1.54x overall (expected: this area's real crime rate (0.29) is much higher than overall (0.25)).

Area	Actual_Rate	Avg_Pred_Prob	Calibration_Gap	High_Flag_Rate	FPR
Civil Line	0.28	0.24	-0.04	0.39	0.34
Civil Line Area A	0.16	0.16	0.00	0.09	0.07
Civil Line Area B	0.19	0.17	-0.02	0.06	0.06
Market Area	0.27	0.28	0.01	0.55	0.51
Market Commercial	0.44	0.40	-0.04	0.90	0.88
Market Residential	0.19	0.20	0.01	0.21	0.17
Waleed Area	0.29	0.26	-0.03	0.51	0.46
Waleed Extension	0.24	0.22	-0.01	0.33	0.29
Waleed North	0.19	0.19	0.00	0.18	0.15

Limitation: The model only learns from RECORDED crimes. If an area has more policing, more crime gets recorded